Using Logistic Regression to Determine U.S. Equity Performance

SSIE 605

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Agenda

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2. Problem Statement
3. Data Capture
4. Data Computations/Clean
5. VIF Analysis
6. PCA Analysis
7. Logisitc Regression & Results
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1. Background

The Securities and Exchange Commission (SEC) requires that publicly traded companies disclose their financial statements on a quarterly basis. These statements offer information about the company's financial health, profitability, and costs. These financial reports can be used to create ratios and other metrics, that are often used by investment analysts to determine if shares in a particular company should be purchased, sold, or left alone. The largest companies in America are tracked by the SP500, and it is assumed that an investor, on average, can do no better than investing in shares of this composite index. It's historical record of returns and overall market exposure make it one of the most attractive investment vehicles on the globe. This projects looks to see if investors can reliably make predictions about it's constituents by using quarterly financial disclosures.

2. Problem Statement

Can we use publicly available financial data from members of the SP500 to successfully determine if a company will outperform, or underperform, the cumulative returns of the SP500 average?

Can we use the current quarter’s information to predict the next quarter’s returns reliably?

Can we use this information to create a portfolio whose returns beat the SP500?

3. Data Capture

Data was sourced from several institutions including

- WRDS (Wharton Research Data Service)
  - UPenn data repository which stores companies' quarterly financials
- FRED (Federal Reserve Economic Data)
  - Federal Reserve Bank which publishes Macroecoonomic Indicators
- EDGAR (Electronic Data Gathering Analysis Retriveal)
  - SEC Repository for Company quarterly filings

Data was pulled via API, and then processed via Pandas in Python

3. Data Catpure

The SP500 Companies were pulled from the Compustat North American Daily All Table [comp_na_daily_all]

The companies' tickers were pulled for each quarter, to get the list of the SP500 members at a particular point in time. Sector was also included

Tickers were used to query the Merged Fundamental Quarterly File Tables [fundq] to get financial information for each company, at each quarter from Q1 2010 until Q4 2025

GDP and inflation measures were pulled as a CSV table from the Financial Reserve Website directly

SP500 performance was pulled from the WRDS repository

3. Data Capture

Variable	**Definition (at end of quarter)**
prccq	Price Close
cshoq	Common shares Outstanding
mkvalq	Market Value
revtq	Revenue
cogs	Cost of Goods Sold
xsgaq	Selling, General and Administrative Expenses
xrdq	Research and Development Expenses

Variable	**Definition**
niq	Net Income
ibq	Income Before Extraordinary Items
piq	Pretax Income
atq	Total Assets
ceqq	Common Equity Total
teqq	Total Stockholders Equity
dlttq	Long Term Debt

Variable	**Definition**
actq	Current Assets
lctq	Current Liabilities
cheq	Cash and Short Term Investments
dpq	Depreciation and Amortization
txtq	Income Taxes
oibdpq	Operating Income before before Depreciation
dlcq	Debt in Current Liabilities

3. Data Computations

From the original data, new variables were created in line with what analysts typically use to evaluate a company’s financial health

Ratio	**Definition (at end of quarter)**
Return On Asset	Net Income / Total Assets
Return on Equity	Net Income / Common Equity Total
Gross Margin	(Revenue - Cost of Goods Sold) / Revenue
Operating Margin	Operating Income before before Depreciation / Revenue
Net Margin	Net Income / Revenue
Asset Turnover	Revenue / Current Assets

Ratio	**Definition (at end of quarter)**
Debt to Equity	Long Term Debt / Common Equity Total
Revenue Growth	(Revenue - Revene Lagged) / lagged revenue
Net Income Growth	(Net Income - Net Income Lagged) / Net Income Lagged
Price to Earning	Price / Income Before Extrordinary Expenses / Common shares Outstanding
Book to Market	Common Equity Total / Market Value
Current Ratio	Current Assets / Current Liabilities

3. Data Computations

Delta data was taken to compare Quarter over Quarter Performance

A companies current financial snapshot was used to determine if they were an outperformer for the final quarter

An outperformer is a company that performed *better* than the average market return for the same time period

Outperformers were labelled ‘1’, underperformers as a ‘0’

ticker	comnam	exchcd	siccd	quarter	quarterly_return	spy_quarterly_return	outperformer_quarterly	quarter_str	year	q
A	AGILENT TECHNOLOGIES INC	1	3825	2010Q1	0.106855	0.048722	1	2010Q1	2010	1
AAPL	APPLE INC	3	3571	2010Q1	0.115161	0.048722	1	2010Q1	2010	1
ABBV	ABBVIE INC	1	2834	2013Q1	0.111474	0.100268	1	2013Q1	2013	1
ABNB	AIRBNB INC	3	9999	2021Q1	0.280246	0.057726	1	2021Q1	2021	1
ABT	ABBOTT LABORATORIES	1	2834	2010Q1	-0.016891	0.048722	0	2010Q1	2010	1
XYL	XYLEM INC	1	2431	2011Q4	0.074895	0.111523	0	2011Q4	2011	4
YUM	YUM BRANDS INC	1	5812	2010Q1	0.102811	0.048722	1	2010Q1	2010	1

3. Data Cleaning

To clean the data, we first winsorized, and then standardized to mitigate the effects of atypical results (outliers)

NAN's were also dropped which took the training dataset from 21285 rows to 16589 rows

5. VIF Analysis

To mitigate collinear predictors, the variance inflation factor was calculated for all the predictors.

- Variables with VIF’s >2.5 were removed one at a time until none were left
- 20 Variables out of the original 24 were kept
- 'roe', 'gross_margin', 'op_margin', 'asset_turnover', 'current_ratio', 'debt_to_equity', 'rev_growth', 'ni_growth', 'pe_ratio', 'book_to_market', 'gdp_growth', 'inflation', 'delta_roe', 'delta_gross_margin', 'delta_net_margin', 'delta_asset_turnover', 'delta_current_ratio', 'delta_debt_to_equity', 'delta_pe_ratio', 'delta_book_to_market'

6. PCA Analysis

PCA was also performed to condense the amount of predictors deemed necessary and further remove the effects of multi-collinearity

12 Principal components were ultimately kept to maintain 80% variance spread

5 Components explain ~46% of Variance

Number of Variables reduced by half

	PC1	PC2	PC3	PC4	PC5
roe	-0.262899	-0.234864	-0.420419	-0.021016	-0.164524
gross_margin	-0.234431	0.520552	-0.178525	-0.117226	-0.079833
op_margin	-0.287513	0.456514	-0.125299	-0.137990	-0.075212
asset_turnover	0.078916	-0.352394	0.154814	0.166886	-0.107535
current_ratio	-0.071214	0.252363	-0.061612	-0.056732	0.007302
debt_to_equity	-0.125429	-0.278715	-0.430184	-0.028188	-0.224704
rev_growth	-0.369397	-0.072816	0.368712	0.062706	-0.260504
ni_growth	-0.356581	-0.099685	0.105818	0.062660	0.355848
pe_ratio	-0.017518	0.162385	-0.144207	0.661489	0.049150
book_to_market	0.110872	-0.173606	0.102219	-0.098844	0.288439
gdp_growth	-0.148228	-0.014681	0.150066	0.113271	-0.286399
inflation	-0.113032	0.009277	0.080542	0.095952	-0.304686
delta_roe	-0.316347	-0.245649	-0.267466	-0.020647	0.107860
delta_gross_margin	-0.307352	-0.004040	0.099351	-0.024738	0.240019
delta_net_margin	-0.398916	-0.113614	0.077943	0.037468	0.389780
delta_asset_turnover	-0.296909	-0.077738	0.357842	0.072139	-0.334388
delta_current_ratio	-0.040757	0.026892	-0.051122	-0.009025	0.230351
delta_debt_to_equity	-0.060528	-0.194895	-0.344758	-0.051237	-0.117830
delta_pe_ratio	0.027944	0.099200	-0.129077	0.668261	0.088997
delta_book_to_market	-0.085692	-0.000672	0.042324	-0.027441	0.189681

7. Logistic Regression (Preliminary)

Logistic Regression was next trained on the dataset spanning all companies and all quarters from 2010 to 2024

The following the preliminary analysis was obtained

— TABLE 8: PCA COMPONENT COEFFICIENTS & SIGNIFICANCE —

Component	Coeff	Odds Ratio	95% CI	Z-Stat	p-Value	Significant?
PC1	-0.0486	0.9525	[0.934,0.971]	-4.956	0	*** p<0.01
PC2	0.0016	1.0016	[0.981,1.023]	0.151	0.8804	No
PC3	0.0272	1.0275	[1.005,1.051]	2.377	0.0174	** p<0.05
PC4	-0.0031	0.9969	[0.973,1.021]	-0.251	0.8022	No
PC5	0.0025	1.0025	[0.977,1.028]	0.194	0.8465	No
PC6	-0.0709	0.9316	[0.907,0.957]	-5.115	0	*** p<0.01
PC7	0.0435	1.0444	[1.015,1.074]	3.013	0.0026	*** p<0.01
PC8	-0.0168	0.9833	[0.956,1.012]	-1.153	0.2489	No
PC9	0.0111	1.0111	[0.982,1.042]	0.734	0.4631	No
PC10	-0.0353	0.9654	[0.936,0.996]	-2.214	0.0269	** p<0.05
PC11	0.018	1.0181	[0.984,1.053]	1.037	0.2996	No
PC12	0.0089	1.0089	[0.975,1.044]	0.506	0.6128	No

— CONCLUSION —

Prediction : Q[t] ratios → Q[t+1] outperformance (forward-looking)
Feature set : 14 base ratios + 10 QoQ delta features = 24 total
Overall model : ✓ SIGNIFICANT
AUC : 0.5403
Pseudo R^2 : 0.0032
Significant PCA components (5/12): PC1, PC3, PC6, PC7, PC10

7. Logistic Regression (Preliminary) - Confusion Matrix

```
── TABLE 7: CONFUSION MATRIX ──────────────────────────────────
  Cutoff = 0.5

                                Predicted: Under (0)  Predicted: Over (1)
  Actual: Underperformer (0)    TN=3,182              FP=4,935
  Actual: Outperformer  (1)     FN=2,808              TP=5,664

  Sensitivity (Recall)  : 66.9%  → correctly identified outperformers
  Specificity           : 39.2%  → correctly identified underperformers
  Precision             : 53.4%  → of predicted outperformers, truly so
  Overall Accuracy      : 53.3%
```

7. Preliminary Results

53.3% Accuracy was obtained, which is hardly better than random, can we do better?

Subsequent Analysis sought to bifurcate the data according to Market Cap and Sector to see the effect it had on model performance

Companies were broken into small, medium, large Market Capitalizations to determine if the model could be improved

MARKET CAP THRESHOLDS

Group	Market Cap Range	Observations
Small Cap	< $18B	7,048 obs
Mid Cap	$18B – $43B	7,225 obs
Large Cap	> $43B	7,012 obs

7. Logistic Regression

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8. Segmentation

Sector	**Observations**	**Accuracy**
Communication Services	428	60.7
Consumer discretionary	2668	54.1
Consumer Staples	1030	55.4
Energy	261	60.5
Financial	4108	57.4
Health Care	1126	56.0
Industrial	4183	53.6
Information Technology	2240	54.7
Materials	2781	56.3
Real Estate	424	59.9
Utilities	1757	57.6

9. 2025 Testing Set

To test performance of the model, the 2025 dataset was usedand the outperformers for a period of 1 year were selected. The predicted underperformers were not as investment vehicles. By using a bench mark of $10,000 portfolio, and allocating this proportionally by market cap of all outperformers, the team was able to compare the quarterly performance of the model against the same amount of money invested into the SP500.

The results were as follows:

10. Results

While segmenting the data by Market Cap and Sector slightly boosted the performance of the model, the team concludes that the Efficient Market Theory holds for this analysis, and that quarterly financial health should not be used to

10. Conclusion

The team DOES OR DOES NOT support the efficient market hypothesis. Quarterly financial statements DO/DO NOT yield a better result than passive investing into a large market index.

8. Segmentation - Market Cap

When breaking up the companies by Market Cap, the following Accuracy scores were obtained:

Market Cap	**Model Accuracy**
Large Cap	54.2%
Mid Cap	52.8%
Small Cap	54.5%

Breaking up the data by Market Cap did not change the results much. Performance was onl y slightly improved for Large and Mid Cap

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Several Variables were computed from the raw data from WRDS. These were the financial ratios used to determine a companies financial health and included computations around their assets, liabilities, capital expenditures, share price, and market capitalization.

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Success factors

Make a claim about what your company needs to do to succeed

Highlight a key attribute of successful brands in your market or industry.

Competitor name

Describe an example of this success factor in action. Explain how it has contributed to the competitor’s business.

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Describe an example of this success factor in action. Explain how it has contributed to the competitor’s business.

State the purpose of this presentation or meeting.
Explain why the content is relevant to your audience.
For example, your objective might be to better understand the competitive landscape to recognize business opportunities.

Objective

State the purpose of this presentation. For example, you might want to understand the competitive landscape to identify opportunities for your business.

How we conducted our analysis

If you repeated steps or iterated on the process, change the conclusion step to show what you repeated.

STEP 1 → STEP 2 → STEP 3 → STEP 4 → CONCLUSION

STEP 1

Walk your audience through your process.

STEP 2

At each step, briefly explain the actions carried out by the team.

STEP 3

For example, they might have identified competitors, or defined analysis criteria.

STEP 4

Describe how they interviewed stakeholders, tested with users, and collected industry data.

CONCLUSION

Eventually, the team arrived at the analysis that you'll be presenting in the slides ahead.

How we conducted our analysis

1. Identified competitors

Split your research methodology into phases. For example, as your first step, you might have identified competitors. Walk your audience through this process. What criteria did you use? What kind of competitors did you focus on?

2. Gathered data

After identifying your competitors, the next step might have been gathering data. If so, what sources did you consult? Did you collect industry reports and customer reviews? Did you conduct surveys and interviews?

3. Analyzed and interpreted data

Finally, what did you do with all this information? How did you analyze and interpret it? Preview the results of your competitive analysis, which will be presented on the following slides.

Challengers

List competitors who currently dominate the market, but are not prepared for the future.

Leaders

List competitors who both dominate the current market and are well-positioned for the future.

List competitors who are successful within their niche, but have no clear plan or roadmap.

Niche players

List competitors who have a clear vision of the future, but have yet to fulfill their business potential.

Visionaries

Competitor landscape

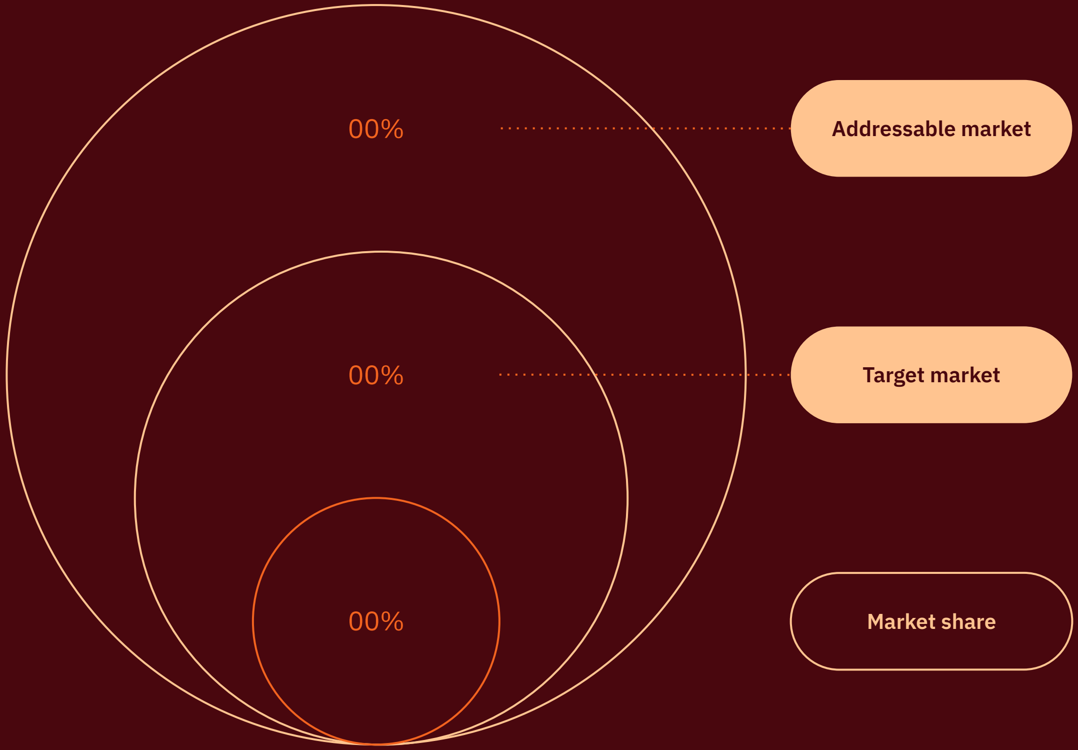
Identify and classify your competitors. Place them into quadrants based on how successful they are now — and how prepared they are for the future.

If you add a diagram to this slide, explain what it shows. Help your audience make sense of the information.

Competitor landscape

Map your competitors based on price and quality, using a brand positioning graph

If you add a diagram to this slide, explain what it shows. Help your audience make sense of what they're seeing.



Market size

Your market share, in context

If you add a diagram to this slide, explain what it shows. Help your audience make sense of what they're seeing.

Market trends

Share your vision about where the market is heading, based on the latest trends

01

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

02

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

03

Identify a market trend

Explain why you consider this to be a current or future trend. Support your case with sales data, user interviews, media coverage, industry reports, etc.

Market trends

1. Explain a current or upcoming market trend

Explain why this is an important trend, how it'll impact your business, and how your competitors are responding to the challenge.

Share the latest sales data or industry metrics that support or signal this trend.

Metric description

00

Metric description

00

Present a case study

Show evidence of the trend. Mention a case study, summarize a report, etc.

LINK TO FULL REPORT

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Metric description

00

Competitive analysis

	YOUR COMPANY	COMPETITOR 1	COMPETITOR 2
PRODUCT OR SERVICE	●	-	-
PRODUCT OR SERVICE	●	-	●
PRODUCT OR SERVICE	●	-	●
PRODUCT OR SERVICE	-	●	●
PRODUCT OR SERVICE	●	-	●

Success factors

Now you've reviewed the results of the analysis, write a memorable conclusion that sets up your key takeaways.

Summarize a key takeaway. Refer to the concrete findings you presented in the previous slides.

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Key takeaways

Summarize a key takeaway. Reference the findings you presented in the previous slides.

What we found

1. List the data, trends, and factors that support this takeaway
2. Restate your most important findings
3. List any additional details you want your audience to remember

00

Metric description

00

Metric description

A few words about a **relevant trend** that illustrates and supports this takeaway.

Logo

Logo

Main competitors

Next steps

Establish action items based on your key takeaways

Recap a key takeaway → Make this takeaway actionable. Suggest what your company or organization can do in response to your analysis.

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Any questions? Ask away!

Give your audience space to resolve immediate questions or concerns.

Thank you!

Encourage your audience to provide feedback and continue the conversation at a later date.

Contact
name@example.com